

Boundary-spanning demand-side search and radical technological innovations in China

BSDSS and
radical
technological
innovations

The moderation of innovation appropriability

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Received 14 April 2016
Revised 9 September 2016
7 June 2017
Accepted 12 June 2017

Abstract

Purpose – The purpose of this paper is to explore how boundary-spanning demand-side search (BSDSS) fuels radical technological innovations as well as how innovation appropriability moderates this relationship. In particular, based on Teece's (1986) argument regarding the appropriability of innovation, the authors divide factors to influence innovation appropriability into two types: external institution related and internal capability related.

Design/methodology/approach – This study employs a survey methodology. Specifically, the authors collected a sample composed of 150 high-tech manufacturing Chinese firms.

Findings – Results show that BSDSS has a positive effect on radical technological innovations. Further, the authors find that dysfunctional competition and political ties negatively moderate the main effect, whereas firms' legal and IPRs protection capabilities positively moderate the main effect.

Research limitations/implications – One major limitation is that the findings are based on data derived from Chinese firms, which may limit the generalization of the findings.

Practical implications – The findings suggest that firms competing Chinese market, both Chinese and foreign firms, should actively leverage BSDSS to boost radical technological innovations. Chinese firms should pay attention to the negative roles of appropriability hazards originating from external institutional environment. Foreign firms in Chinese market should be cautious on potential dysfunctional competition from local competitors, such as imitation and intelligence property violation, and enhance appropriability through building internal capabilities, such as legal and IPRs capabilities.

Originality/value – The study highlights the crucial roles of BSDSS in radical technological innovations, as well as the moderating roles of innovation appropriability. These results provide new insights into the drivers of radical technological innovations.

Keywords China, Boundary-spanning demand-side search, Innovation appropriability, Radical technological innovations

Paper type Research paper

Introduction

Radical technological innovation, that represents state-of-the-art technological advance (Benner and Tushman, 2003; Chandy and Tellis, 1998), is a cornerstone in sustaining competitive advantages and growth (Kleinschmidt and Cooper, 1991; Tellis *et al.*, 2009).

This research was supported by the National Natural Science Foundation of China (Nos 71372026, 71421002).



Management Decision
Vol. 55 No. 8, 2017
pp. 1749-1769
© Emerald Publishing Limited
0025-1747
DOI 10.1108/MD-04-2016-0236

However, developing radical technological innovations is not only difficult but also risky (Kleinschmidt and Cooper, 1991; Tellis *et al.*, 2009). Innovations are intrinsically about identifying and using new and unique knowledge to achieve technological breakthrough (Van de Ven, 1986). Thus, recent literature, adopting knowledge-based view, pays attention to knowledge in radical technological innovations (Miller *et al.*, 2007; Subramaniam and Youndt, 2005; Zhou and Li, 2012; Zhou and Wu, 2010). Especially, in the increasingly popular of the “open innovation” model (Chesbrough, 2003), the boundary-spanning search, the proactive process to search and identify of new knowledge from wide range, is a central part in developing innovations (Rosenkopf and Nerkar, 2001). External knowledge from boundary-spanning search helps overcome core rigidities (Leonard-Barton, 1992) and competency traps (Levitt and March, 1988). Thus, boundary-spanning search has been identified as an important driver of radical technological innovations (Laursen and Salter, 2006; Phene *et al.*, 2006; Rosenkopf and Nerkar, 2001).

Since technology-push perspective has been influential in innovation-related literature, many studies focus on the technology aspect of supply side search (e.g. Ahuja and Lampert, 2001; Katila and Ahuja, 2002; Laursen and Salter, 2006; Nerkar, 2003). However, radical technological innovations’ risk from demand side is also considerably high (Christensen and Bower, 1996; Lee and O’Connor, 2003). Thus, the emerging demand-side approach has highlighted the critical roles of demand-side search of market knowledge in innovations (Adner and Levinthal, 2001; Priem *et al.*, 2012). The boundary-spanning demand-side search (BSDSS), defined as “the effort devoted to the discovery of new insights in own and allied markets regarding market structures and segments, product use and substitution patterns, and customer preferences and needs” (Sidhu *et al.*, 2007, p. 21), will facilitate firms to deeply understand market trends, and direct firms to develop advanced technologies that best fit with customers’ needs (Cooper and Inoue, 1996; Nerkar and Roberts, 2004). Thus, not only can BSDSS serve as a key driver of radical technological innovations, but also can effectively reduce the risk of developing radical technological innovations (Priem *et al.*, 2012).

BSDSS is important for firms ambitious in acquiring cutting edge technologies through radical innovations, especially firms in emerging economies like China. On the one hand, Chinese firms, unlike their counterparts in developed economies, have fallen prey to entrenched technological disadvantages, so that it is hard for them to create disruptive technologies through technology push. On the other hand, one noteworthy advantage for Chinese firms to develop radical technological innovations is the massive opportunities in Chinese market. Huge market will facilitate Chinese firms to discover innovation opportunities and to commercialize new ideas fast (Zhou and Li, 2012; Zhou and Wu, 2010). Unfortunately, as a world factory, Chinese firms have long anchored their attention on efficiency and cost, rather than on listening to their customers (Park and Luo, 2001). Hence, the demand-side approach may yield new strategic mindsets for Chinese firms, which are ambitious in radical technological innovations. However, given the dominance of technology-push explanations, only limited research has explored the effect of BSDSS on innovations in general, rather than on radical technological innovations in specific (e.g. Sidhu *et al.*, 2007). Thus, based upon the practical urgency and theoretical void, we propose our first question: how will BSDSS affect radical technological innovations for firms in China?

While BSDSS has potential to discover innovation opportunities and reduce market risks, innovation appropriability, which refers to firms’ abilities to capture the profits generated by their innovations (Teece, 1986), must be considered when studying the link between BSDSS and radical technological innovations. This is because appropriability hazards will deeply hurt innovation benefits (Li and Atuahene-Gima, 2001; Zhang *et al.*, 2007), and thus abate firms’ enthusiasm to leverage BSDSS for radical technological innovations. Appropriability hazards originate from the pervasive presence of behavioral uncertainty, combined with the difficulties of specifying intellectual property rights (Gulati and Singh, 1998; Pisano, 2006; Teece, 1986).

In a weak appropriability regime where appropriability hazards are pervasive, new technologies are easily to be copied by competitors, and thus innovating firms may fail to obtain economic returns (Zhang *et al.*, 2007). In this case, appropriability will determine whether firms actively leverage the innovation opportunities discovered by BSDSS. In particular, comparing with counterparts in developed economies, Chinese firms suffer more salient appropriability hazards because of lacking well-established legal frameworks to protect property rights (Hoskisson *et al.*, 2000; Li and Atuahene-Gima, 2001). However, no research so far, to our best knowledge, has empirically investigated the contingent effects of innovation appropriability on the relationship between BSDSS and radical innovations in a Chinese context. Thus, our second question is: how does appropriability moderate the relationship between BSDSS and radical technological innovations in China?

We contribute to the emerging demand-side approach on innovations by clarifying how BSDSS will drive radical technological innovations in China (Priem *et al.*, 2012). Moreover, to highlight the moderating effects of appropriability on the link between BSDSS and radical technological innovations, this study integrates two separated streams literature: innovation search and innovation appropriability.

This study also has robust practical implications for both Chinese firms and foreign firms that have been interested in Chinese market. To Chinese firms that have been confronting intense competition from foreign counterparts, “listening to market” is especially vital due to their technological disadvantages. Moreover, we suggest that Chinese authorities should enhance legal institutions to protect innovative efforts, such as laws and regulations (Li and Atuahene-Gima, 2001). In addition, since Chinese institutions are still at nascent stage, local competitors may compete unfairly against foreign firms, such as imitating their innovation technologies. To foreign firms, we suggest that those implementing radical technological innovations in China should find creative ways to protect their innovation efforts, such as preparing for law knowledge and building legal and IPRs capabilities.

Literature review

Radical technological innovations

Technological innovations have two types: incremental technological innovations and radical technological innovations (e.g. Danneels and Kleinschmidt, 2001; Page and Schirr, 2008). Incremental technological innovations refer to minor changes or simple improvements in technologies, whereas radical technological innovations represent novel, unique, or state-of-the-art technological advances (Benner and Tushman, 2003; Chandy and Tellis, 1998). Since radical technological innovations adopt new and advanced technologies, they may fundamentally change the technological trajectories and alter the consumption patterns (Zhou and Li, 2012). Thus, while creating great value for firms in terms of growth, sustainability, and competitive advantage (Chandy and Tellis, 1998; Sorescu *et al.*, 2003; Tellis *et al.*, 2009; Wind and Mahajan, 1997), radical technological innovations are highly risky to pursue because of technical and market uncertainty (O'Connor and Veryzer, 2001; Zhou and Li, 2012).

As market competition is becoming increasingly fierce in Chinese market, radical technological innovation has received great attention of Chinese firms (Atuahene-Gima, 2005; Zhang and Li, 2010; Zhou, 2006). Chinese firms lack core technologies, for a long time their technology development depends on absorbing and adapting knowledge from foreign counterparts. It is time consuming to build a base of technological knowledge that is sufficient to achieve technological breakthrough. Thus, Chinese firms have confronted big obstacles to develop radical technological innovations, which are pushed by technological breakthrough.

The antecedents of radical technological innovations

Extensive studies have explored the antecedents of radical technological innovations, and identified a list of organizational components that influence the development of radical innovations: firm resources (Voss *et al.*, 2008), market power (Sorescu *et al.*, 2003), firm leadership (Kashmiri *et al.*, 2017; Strese *et al.*, 2016), organizational cultures (Chandy and Tellis, 1998; Tellis *et al.*, 2009), organizational structure (Ettlie *et al.*, 1984; Leifer *et al.*, 2001; O'Connor and Demartino, 2006), and strategic orientations (Zhou *et al.*, 2005).

Besides above-mentioned views on radical innovations, another stream perceives radical innovations from a knowledge-based view. Because innovations are intrinsically about identifying and using new knowledge to achieve technological breakthrough (Van de Ven, 1986), recent innovation literature has adopted the knowledge-based view (Miller *et al.*, 2007; Subramaniam and Youndt, 2005; Zhou and Li, 2012; Zhou and Wu, 2010). For example, Zhou and Li (2012) have examined how existing knowledge base and knowledge integration jointly affect radical innovations.

In particular, with the increasingly popular of the “open innovation” model (Chesbrough, 2003), radical innovations becomes a function of the external knowledge acquisition (Phene *et al.*, 2006), because external knowledge helps overcome core rigidities (Leonard-Barton, 1992) and competency traps (Levitt and March, 1988). Therefore, one influential stream has emphasized the critical roles of boundary-spanning search, which is the proactive process to search and identify new knowledge from a wide range of external sources (Rosenkopf and Nerkar, 2001). Because radical technological innovations require novel technological knowledge, this research has focused on the technology aspect of supply side search, and found the positive role of supply side search in radical technological innovations (e.g. Ahuja and Lampert, 2001; Laursen and Salter, 2006; Maggitti *et al.*, 2013; Phene *et al.*, 2006).

In addition, due to the high uncertainty, appreciating the value of disruptive technologies for marketplace growth is an important innovation challenge (Reid and De Brentani, 2010). Because technological discontinuity often encounters adoption barriers and market failure (Aarikka-Stenroos and Lehtimäki, 2014; Chiesa and Frattini, 2011; Lee and O'Connor, 2003), radical technological innovations derived from demand side are risky (Christensen and Bower, 1996; Lee and O'Connor, 2003). Successful radical technological innovations have to meet the customers' needs (Fleming, 2002; Nerkar and Roberts, 2004). The key is to link advanced technologies to market opportunities of the future, so that radical technological innovations may be effectively guided and motivated (O'Connor, 1998; Reid and De Brentani, 2010).

Chinese firms do enjoy one advantage in developing radical technological innovations: huge market. Considering Chinese firms' three merits: massive market, low cost, and flexibility, they can transform their ideas into commercial products more quickly and at lower cost than their counterparts in developed countries (Zhou and Li, 2012). In fact, several studies have found empirical evidence that supporting market will drive radical technological innovations in China. For example, Zhou *et al.* (2005) have found that a market orientation facilitates technology-based breakthrough innovations for firms in China. Zhou and Li (2012) have found that high technology firms in China are more likely to achieve radical technological innovations when integrating potential market knowledge with their understandings over current segments. Cai *et al.* (2014) have found that responsive market orientation promotes radical technological innovations for Chinese new ventures.

Thus, the rapidly growing demand-side approach has highlighted the critical roles of demand-side search in driving innovations (Adner and Levinthal, 2001; Priem *et al.*, 2012). Through searching for demand-side informations, firms can better understand customers' needs, identify potential market opportunities, and thus successfully launch new products that best satisfy customers. For instance, Sidhu *et al.* (2007) have found that in addition to

supply side search, BSDSS improves a firm's innovation performance. Unfortunately, as the world factory, Chinese firms have paid little attention to demand-side search to boost radical technological innovations (Park and Luo, 2001).

Although researchers have started to investigate the link between market and firm radical technological innovations, few have clarified, why, how, and when BSDSS will affect radical technological innovations for firms in China. In following parts, we will start to fill this theoretical void.

Hypotheses development

Effects of BSDSS on radical technological innovations

Researchers maintain that overemphasis on markets could lead to trivial innovations and lower the firm's radical technological innovations (Christensen and Bower, 1996; Slater and Narver, 1998; Zhou and Li, 2012). However, we argue that BSDSS can improve firms' radical technological innovations, because it spans market niches occupied by rivals, and helps recognize customers' unmet needs or target potential market opportunities (Sidhu *et al.*, 2007). Such search is likely to sharpen firms' awareness and knowledge over the needs and preferences of current as well as potential customers, provide insights into customers' product use and substitution patterns, and give exposure to new-to-the-firm solutions to issues associated with the product offering (Cooper and Inoue, 1996; Kohli and Jaworski, 1990). Identifying market opportunities of the future can direct and motivate firms' advanced technologies development (O'Connor, 1998; Reid and De Brentani, 2010).

In addition, radical technological innovations driven by demand-side knowledge will reduce firms' failure risk, because radical technological innovations are more feasible to be recognized and accepted by customers (Zhou *et al.*, 2005). Thus, not only can BSDSS provide opportunities for firms' radical technological innovations, but also can enhance firms' motivations to invest into radical technological innovations.

To be specific, we expect that BSDSS will drive radical technological innovations for firms in China. First, the technologies of Chinese firms are still at initial phase, and thus radical technological innovations are more difficult to be pushed by technological breakthrough. Meanwhile, China's growing market provides a number of market opportunities, which will facilitate Chinese firms to breed radical technological innovations. Furthermore, radical technological innovations driven by demands will substantially reduce technological and market uncertainty around them, which is critical for pursuing radical technological innovations. In all, since BSDSS can deepen Chinese firms' understanding over market needs, their intention and motivation to pursue radical technological innovations will be boosted. Therefore, we propose that:

H1. BSDSS has a positive effect on radical technological innovations in China.

Innovation appropriability as contingency

While BSDSS has potential to discover innovation opportunities and reduce market risk, innovation appropriability, which refers to firms' abilities to capture profits generated by their innovations (Teece, 1986), must be considered when studying the role of BSDSS in radical technological innovations. This is because appropriability hazards may hurt innovation benefits (Li and Atuahene-Gima, 2001; Zhang *et al.*, 2007), and thus will abate firms' enthusiasm to leverage BSDSS for radical technological innovations. Whether firms can capture profits from their innovations is an important issue, especially for radical technological innovations that require substantial resources (Li and Atuahene-Gima, 2001; Sorescu *et al.*, 2003). Firms hardly can gain benefits from radical technological innovations if their rivals easily copy the ideas and knowledge. If firms cannot protect and reap benefits from radical technological innovations, they have no willingness to undertake such risky

activities (Hurmelinna-Laukkanen *et al.*, 2008; Ritala and Hurmelinna-Laukkanen, 2013). It is necessary of building an appropriability regime, which protects against imitation and enables getting returns on investments in radical technological innovations (Teece, 1986). Thus, firms' incentives to leverage these opportunities from BSDSS will largely depend on their perceived innovation appropriability.

In particular, appropriability hazards are pervasive and fatal for firms in China. Prior research has noted that comparing with developed markets, appropriability hazards are more salient in Chinese market, which suffers a loose appropriability regime and an incomplete legal framework to protect intellectual property rights (Hoskisson *et al.*, 2000; Li and Atuahene-Gima, 2001; Zhang *et al.*, 2007). Specifically, in a loose appropriability regime, because firms are subject to spillovers to their competitors, they cannot retain the profits from innovations (Gulati and Singh, 1998). Therefore, appropriability hazards will hurt innovation benefits, and thus will abate Chinese firms' enthusiasm to launch innovations (Li and Atuahene-Gima, 2001).

Several empirical studies have supported above arguments. Li and Atuahene-Gima (2001) find that because in such environment the intellectual property rights resulting from innovation may go unprotected, making innovation a highly risky and less profitable strategy, dysfunctional competition negatively moderates the relation between innovation and performance in China. Zhang *et al.* (2007) examine how appropriability hazards moderate the relationship between R&D intensity and international joint venture performance in Chinese market. Combining these two factors of BSDSS and appropriability together can provide insights into how well Chinese firms can create radical technological innovations.

Appropriability is composed of two elements: the nature of knowledge (tacit vs codified) and the strength of the legal protection (such as patents, copyrights, trademarks) (Teece, 1986). According to Teece (1986), appropriability regime is strong when knowledge is difficult to imitate (i.e. the tacit nature of knowledge) and/or it can be legally protected with IPRs. Strong appropriability can effectively prevent rivals from viciously imitating firms' innovations, thereby reducing firms' uncertainties of reaping profits from such initiatives (Dechenaux *et al.*, 2008; Teece *et al.*, 1997). While the tacit nature of knowledge is a critical element for appropriability, in this study, we mainly focus our attention on external institutional environments and internal capabilities.

On the one hand, the relatively underdeveloped legal and financial institutions in China will greatly increase the importance of external institutional environments, such as dysfunctional competitions and political ties. Prior studies have argued that dysfunctional competitions and political ties can greatly influence innovation appropriability. For example, Li and Atuahene-Gima (2001) highlight that dysfunctional competition and government support will affect the effectiveness of Chinese firms' innovation strategy. On the other hand, previous research has pointed that firms in emerging economies with weak appropriability regimes can also leverage internal capabilities (Hurmelinna-Laukkanen *et al.*, 2008), such as legal capability in general and IPRs capability in particular to safeguard firms' innovations (Keupp *et al.*, 2010; Maine and Garnsey, 2006). For example, Su *et al.* (2013) find that firms in China can properly leverage their capabilities, especially legal capability, to reap profits from launching innovations.

Thus, based upon Teece (1986), we categorize firms' innovation appropriability into two types: appropriability related with external institutional environments and appropriability related with internal capabilities. Previous research mainly focuses on how appropriability affects the relationship between innovations and performance. We, however, drawing on Teece's (1986) appropriability, argue that both external institutional environments including dysfunctional competition and political ties and internal capabilities including legal and IPRs capabilities will moderate the link between BSDSS and radical technological innovations. Our efforts are new in innovation literature.

Moderating roles of appropriability related with external institution environments

Dysfunctional competition and political ties represent two types of institution environments in China (Li and Atuahene-Gima, 2001). We first suggest that the association between firms' BSDSS and radical technological innovations will be weakened by dysfunctional competition. Dysfunctional competition refers to the extent to which the competitive behaviors in a market are opportunistic, unfair, or even unlawful (Li and Atuahene-Gima, 2001). High level dysfunctional competition will discourage firms from employing the opportunities, knowledge, and information derived from BSDSS, thereby negatively moderating the main effect.

Specifically, in China, whose legal institutions are relatively underdeveloped, firms in such transitional economies may have motivation to engage in various dysfunctional competitions, including opportunistic and unlawful behaviors (Nee, 1992; Peng and Heath, 1996). Li and Atuahene-Gima (2001) have argued that patent and copyright violations, broken contracts and agreements, and unfair competitive practices have been widespread in China. As a consequence, firms in such institutional environments replete with dysfunctional competitions, may concern that their radical innovations will go unprotected, thereby making radical technological innovation initiatives highly risky. Therefore, dysfunctional competition may discourage firms from leveraging the opportunities derived from BSDSS to implement radical technological innovations. Therefore, we propose that:

- H2.* In China, the relationship between BSDSS and radical technological innovations will be negatively moderated by perceived dysfunctional competition.

On the contrary, we propose that the link between firms' BSDSS and radical technological innovations will be strengthened by firms' political ties. Political ties, as an institutional context, refer to the extent to which the firms are embedded with authorities, government, and officials (Peng and Luo, 2000). Since firms with political ties will acquire more institutional support and protection from government, such firms may be bolder when deciding to implement risky strategies, such as radical technological innovations (Li and Zhang, 2007). Hence, high level political ties may encourage firms to employ the opportunities, knowledge, and information derived from BSDSS, thereby positively moderating this relationship.

Specifically, in emerging economies like China, government's supports derived from political connections particularly vital for firms (Peng and Heath, 1996). Peng and Luo (2000) have found that since China's formal institutions are highly underdeveloped, political ties can reduce Chinese firms' institutional uncertainty, thereby boosting their performance. Since radical technological innovations will considerably consume resources, strong political ties will provide firms policy information and institutional protection, thereby fueling their motivation to launch radical technological innovations. Therefore, we propose that:

- H3.* In China, the relationship between BSDSS and radical technological innovations will be positively moderated by political ties.

Moderating roles of appropriability related with internal capabilities

Firm's capabilities are the "complex bundles of skills and accumulated knowledge that enable firms to coordinate activities and make use of their assets" (Day, 1994, p. 38). In emerging economies with weak appropriability regimes like China, firms can rely on their internal capabilities to reap the benefits of innovations (Su *et al.*, 2013). Built upon this logic, we argue that the association between Chinese firms' BSDSS and radical technological innovations will be strengthened by firms' internal capabilities to protect innovations.

In this study, we focus on legal and IPRs capabilities that can keep Chinese firms' radical technological innovations from illegal imitation. Legal capability refers to a firm's abilities to deal with legal affairs and to utilize the law to safeguard its interests and advantages

(McGahan and Silverman, 2006). In their study, Su *et al.* (2013) have found that firms' legal capability helps firms reap profits from their innovations. High level legal capability will strengthen Chinese firms' confidence to employ the opportunities and information derived from BSDSS, to launch radical technological innovations. Therefore, we propose that:

H4. In China, the relationship between BSDSS and radical technological innovations will be positively moderated by firms' legal capability.

Similar with legal capability, IPRs capability is also able to strengthen the association between BSDSS and radical technological innovations. IPRs capability refers to a firm's ability to use patents, trademarks, and designs to effectively protect innovation returns. IPRs capability has long been considered as an effective way to protect innovations. For example, Kyläheiko *et al.* (2011) have found that firms' IPRs will drive innovation-based growth. When firms are able to create stronger appropriability of their intellectual property resulting from radical technological innovations, their motivation and intention to launch radical technological innovations based on new market opportunities from BSDSS will be boosted (Hurmelinna *et al.*, 2007; Pisano, 2006). Therefore, we propose that:

H5. In China, the relationship between BSDSS and radical technological innovations will be positively moderated by IPRs capability.

Methodology

Sampling and data

To test our hypotheses, we collected a sample of high-tech manufacturing firms in the China-Singapore Suzhou Industrial Park. Such park is one of the most innovative industrial parks in China, and firms in the park actively engage in innovation initiatives, especially radical technological innovations. Since our primary focus is fastened on how BSDSS will propel firms' radical technological innovations, a sample of high-tech manufacturing firms that comprehensively entail radical technological innovations will act as an appropriate empirical context to test our hypotheses.

Specifically, we employed questionnaire survey method to collect research data. The survey is conducted in 2012 with three steps. First, we developed a questionnaire based on previous studies and used the two-way translation method to ensure accuracy. We also modified it through consulting with executives as well as conducting a pilot test. To reduce respondents' workload and improve accuracy, we divided the questionnaire into two parts. Part I includes the questions regarding the firm's basic information, radical technological innovations, BSDSS, external environments, and so on. Part II is made up of questions about IPRs capability, legal capability, and others. And, in every firm we chose two top managers to answer each part independently.

Second, with the help of Administrative Committee of Suzhou Industrial Park, we obtained a directory of manufacturing firms in this park, and we randomly selected 500 firms from the directory. Then, we contacted with those firms, gave a brief outline of the survey, and requested the firms' participation.

Third, we used face-to-face interview method to obtain responses. All interviewers were doctoral candidates and professors in Chinese universities. They were appropriately trained before starting interviews. The training covered background information, interviewing skills, and the exact meaning of each question in the questionnaire. After a special effort to contact with executives in each firm, the interviewers went to the firm to interview. They began by showing a letter that explained the intentions of the survey, promised to keep responses confidential. Then, they required two top managers who are familiar with the firm's strategies and innovation activities to finish the Part I and Part II of one questionnaire independently.

Finally, we obtained 150 usable questionnaires, with a valid response rate of 30 percent. Such response rate is relatively high, especially considering these survey questionnaires were mostly completed by CEOs and top managers whose time is rather precious. In all, our data are appropriate to test our hypotheses.

Sample selection bias

Sample selection bias may be a concern. Of the responding high-tech firms, 37 percent were in electric products and metalworking industries, 30 percent in electronic information industry, 11 percent in pharmaceutical and bioengineering industries, 10 percent in chemical industry, and 8 percent in appliance industry. We compared responding and non-responding firms along major attributes, including firm age and firm size by the *t*-test. All *t*-statistics were insignificant ($p < 0.05$), suggesting non-response bias is not serious. We also compared our sample with a full sample of high-tech firms issued by National Statistic Bureau along major attributes, including firm age and firm size by the *t*-test. All *t*-statistics were insignificant ($p < 0.05$), suggesting sample selection bias is not serious in our study.

Measurement

Unless stated otherwise, we use a five-point scale in the questionnaire items where “1” represents “strongly disagree” and “5” represents “strongly agree.” Based on previous studies (e.g. Chandy and Tellis, 2000; Su *et al.*, 2013), we measured radical technological innovations by five items. BSDSS was measured by six items developed by Sidhu *et al.* (2007). Dysfunctional competition was measured by four items based on previous studies (e.g. Li and Atuahene-Gima, 2001). Political ties were measured by three items based on the research of Peng and Luo (2000). Legal capability was measured by four items based on previous studies (e.g. Su *et al.*, 2013). Based on previous studies (Dolfsma, 2006; Teece, 1986), IPRs capability was measured by four items. Respondents were asked to rate the extent to which the firm can protect their innovations by using patent, copyright, trade secret, and trademark.

Control variables. We controlled three variables accounting for firms’ characteristics: firm age, firm size that is measured by the number of full-time employees, and whether firms are foreign firms (1 = yes, 0 = no). External environment may influence firms’ innovations (Damanpour, 1991). In this study, we controlled two critical components of external environment (i.e. competition intensity and industry growth). We used items adopted from Jaworski and Kohli (1993) and Zhou (2006) to measure competition intensity and industry growth. All measurement items are shown in Table I.

Reliability and validity

The reliability of multi-item constructs was assessed by Cronbach’s α . Typically, 0.70 or higher is considered an adequate α value (Cronbach, 1971). As shown in Table I, all α values are greater than 0.70. This research conducted confirmatory factor analysis (CFA) with the items linked to prior defined constructs to test the validity. The results indicate that the measuring model fit the data well ($\chi^2/\text{df} = 1.554$; CFI = 0.919; IFI = 0.920; NNFI = 0.909; RMSEA = 0.061). All items are loaded significantly on their corresponding latent construct (the lowest *t*-value is 7.03 with the benchmark is 2.0), which provided evidence for convergent validity.

Discriminant validity is assessed by χ^2 difference test for all multi-item constructs in pairs to see if they are distinct from one another (Zhang and Li, 2010). We combined each pair of constructs into a single model and compared its fit to that of a two-construct model. The results of pair-wise tests indicate that in each case the difference in χ^2 value is significant. Thus, our measures have good discriminant validity.

Variables and items	Standardized loading	t-value
<i>Radical technological innovations</i> ($\alpha = 0.906$)		
Rate the degree to which you agree with each of these statements in the last 3 years:		
1. We developed new technologies which were very novel for our industry	0.77	10.92
2. We developed new technologies which were challenging to existing ideas in our industry	0.81	11.59
3. We developed new technologies which were offering new ideas to our industry	0.88	13.18
4. We developed new technologies which were interesting to our customers and competitors	0.81	11.76
5. We developed new technologies which were promoting fresh thinking to our firm	0.80	11.53
<i>BSDSS</i> ($\alpha = 0.873$)		
Rate the degree to which you agree with each of these statements in the last 3 years:		
1. Marketing strategies of companies targeting our customers are closely followed by us	0.66	8.70
2. We have a finger on the pulse as far as changes in the product preferences of our customers are concerned	0.64	8.37
3. Much information is gathered on product preferences of customer groups that we do not currently serve	0.83	11.89
4. Developments in industries that fulfill the same customer need as we do, albeit with a completely different product, are well known to us	0.84	12.14
5. We keep close track of activities of companies that offer complementary products	0.80	11.30
6. We know the product and process innovation efforts of our customers well	0.63	8.17
<i>IPRs capability</i> ($\alpha = 0.743$)		
Indicate the extent to which your firm can effectively protect innovations using the following:		
1. Patent	0.74	9.52
2. Copyright	0.66	8.17
3. Trade secret	0.71	8.97
4. Trademark	0.70	8.75
<i>Legal capability</i> ($\alpha = 0.770$)		
Rate the degree to which you agree with each of these statements in the last 3 years:		
1. We have strong legal support staffs	0.62	7.54
2. Members of the top management team have strong legal backgrounds	0.81	10.46
3. We have stronger abilities to deal with legal affairs than our major competitors	0.71	8.86
4. We emphasize the importance of law in business activities and operations	0.58	7.03
<i>Political ties</i> ($\alpha = 0.869$)		
Rate the degree to which you agree with each of these statements in the last 3 years:		
1. Improving our relationships with government have been important to us	0.78	10.73
2. We take various measures to in building relationships with government	0.92	13.54
3. We have cultivated close connections with government	0.81	11.28
<i>Dysfunctional competition</i> ($\alpha = 0.875$)		
Indicate the extent to which your principal industry has experienced the following in the last 3 years:		
1. Unlawful competitive practices such as illegal copying of new products	0.80	11.30
2. Counterfeiting of your firm's own products and trademarks by other firms	0.75	10.18
3. Ineffective market competitive laws to protect your firm's intellectual property	0.84	12.05
4. Increased unfair competitive practices by other firms in the industry	0.81	11.52
<i>Industry growth</i> ($\alpha = 0.834$)		
Rate the degree to which each of these statements describes your principal industry over the last three years:		
1. There has been high growth in demand in this industry	0.74	9.72
2. This industry offered many attractive opportunities for future growth	0.91	12.66
3. Growth opportunities in this industry have been abundant	0.74	9.79

(continued)

Table I.
Construct
measurement and
confirmatory factor
analysis results

Table I.

Variables and items	Standardized loading	t-value
<i>Competition intensity</i> ($\alpha = 0.764$)		
Rate the degree to which each of these statements describes your principal industry over the last three years:		
1. Price competition is a hallmark of our industry	0.66	7.89
2. Firms in our industry react swiftly to others' competitive actions	0.70	8.59
3. One hears of a new competitive move almost every day	0.66	7.99
4. Competition in our industry is cut-throat	0.67	8.11
<i>Model fit index</i>		
$\chi^2/\text{df} = 1.554$; CFI = 0.919; IFI = 0.920; NNFI = 0.909; RMSEA = 0.061		

Based on the recommendation from Podsakoff *et al.* (2003), both procedural methods and statistical techniques were taken to reduce the possibility of common method variance. Procedurally, we divided the questionnaire into two parts, and collected data from two managers in each firm. In addition, we assured that their answers would be kept confidential and that there were no right or wrong answers. Statistically, we tested it via Harman's one-factor test. Common method variance would result in one general factor accounting for the majority of covariance in the variables. The test found that no general factor was apparent, with the first factor accounting for 20.1 percent of total variance. We also tested the potential common method variance *ex post* with a CFA approach. We assessed a model that linked all items of the dependent and independent variables to a single factor. This model does not fit the data well ($\chi^2/\text{df} = 5.339$; CFI = 0.548; IFI = 0.552; NNFI = 0.518; RMSEA = 0.171). When all items were assigned to their theoretical factors, the model fits the data well. Therefore, common method bias is not likely to be a threat to the findings of this study.

Analysis and results

Table II presents basic statistics. We tested our hypotheses by hierarchical regression analysis. Cohen *et al.* (1983) suggest that it is appropriate to use a hierarchical approach when analyzing multiplicative terms in regression analysis. We mean-centered all variables to minimize the threat of multicollinearity before creating interaction terms (Aiken and West, 1991). We calculated the variance inflation factors (VIFs) statistics to check multicollinearity. The maximum VIF value is 1.2, which is much lower than the cut-off point 10 (Neter *et al.*, 1985).

Table III contains the hierarchical regression results. *H1* posits that BSDSS has a positive effect on radical technological innovations. Model 2 shows that BSDSS is positively related with firms' radical technological innovations ($\beta = 0.301$, $p < 0.001$). Therefore, *H1* is strongly supported.

H2 posits that the relationship between BSDSS and radical technological innovations will be negatively moderated by dysfunctional competition. Model 3 shows that the interactive terms between BSDSS and dysfunctional competition are negatively associated with firms' radical technological innovations ($\beta = -0.244$, $p < 0.001$). Therefore, *H2* also strongly is supported.

H3 argues that the relationship between BSDSS and radical technological innovations will be positively moderated by political ties. Model 4 shows that the interactive terms between BSDSS and political ties are negatively related with radical technological innovations ($\beta = -0.339$, $p < 0.001$). Therefore, *H3* is not supported.

H4 posits that the link between BSDSS and radical technological innovations will be positively moderated firms' legal capability. Model 5 shows that the interactive terms

Table II.
Descriptive statistic
and correlation matrix

Variables	Mean	SD	1	2	3	4	5	6	7	8	9	10
1. Firm age	18.229	22.092										
2. Firm size	3,736.2	14,079.2	0.050									
3. Foreign ownership	1.407	0.493	-0.092	0.105								
4. Competition intensity	3.808	0.683	-0.161*	0.056	0.213**							
5. Industry growth	3.528	0.79	0.036	-0.069	-0.116	0.077						
6. Dysfunctional competition	3.052	0.978	0.052	0.036	0.109	0.241**	-0.061					
7. Political ties	3.753	0.808	0.019	0.084	0.153	0.116	0.220**	0.151				
8. Legal capability	3.402	0.697	0.018	-0.005	-0.117	0.023	0.200*	-0.035	0.193*			
9. IPRs capability	3.884	0.776	0.067	0.008	0.074	-0.104	0.171*	-0.065	0.179*	0.295**		
10. BSDSS	3.753	0.643	0.087	0.037	-0.062	0.225**	0.125	0.107	0.196*	0.205*	0.028	
11. Radical technological innovations	3.531	0.783	0.000	0.027	0.067	0.129	0.260**	0.068	0.249**	0.356**	0.326**	0.319**
Notes: $n = 150$. * $p < 0.05$. ** $p < 0.01$												

Variables	M1	M2	M3	M4	M5	M6
Firm age	-0.035	-0.054	0.043	-0.018	-0.053	-0.037
Firm size	-0.115	-0.071	-0.134****	-0.112	-0.083	-0.002
Foreign ownership	0.040	0.097	0.069	0.067	0.102	0.051
Competition intensity	0.126****	0.072	0.068	-0.168*	0.122	0.139*
Industry growth	0.387***	0.359***	0.384***	0.314***	0.275***	0.245***
BSDSS		0.301***	0.147*	0.359***	0.219**	0.314***
Dysfunctional competition			0.147*			
Political ties				0.237***		
Legal capability					0.225***	
IPRs capability						0.313***
BSDSS × dysfunctional competition			-0.244***			
BSDSS × political ties				-0.339***		
BSDSS × legal capability					0.171**	
BSDSS × IPRs capability						0.129****
<i>n</i>	150	150	150	150	150	150
<i>F</i> -value	2.956**	3.737***	3.262***	3.949***	2.877***	2.897***
<i>R</i> ²	0.199	0.258	0.316	0.390	0.332	0.349
Adjusted <i>R</i> ²	0.132	0.189	0.219	0.292	0.217	0.229

Table III.
Results of regression analysis^a

Notes: ^aStandardized coefficients are reported here. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.10$

between BSDSS and legal capability are positively related with radical technological innovations ($\beta = 0.171$, $p < 0.01$). Therefore, *H4* is supported.

H5 argues that the relationship between BSDSS and radical technological innovations will be positively moderated by firms' IPRs capability. Model 5 shows that the interactive terms between BSDSS and IPRs capability are positively associated with firms' radical technological innovations ($\beta = 0.129$, $p < 0.10$). Therefore, *H5* is supported.

To facilitate interpretation of the findings, we plotted the moderating effects of dysfunctional competition, political ties, legal capability, and IPRs capability on the relationship between BSDSS and radical technological innovations in Figures 1-4. All figures are consistent with the regression results. For example, Figure 1 shows that the relationship between BSDSS and radical technological innovations becomes less positive

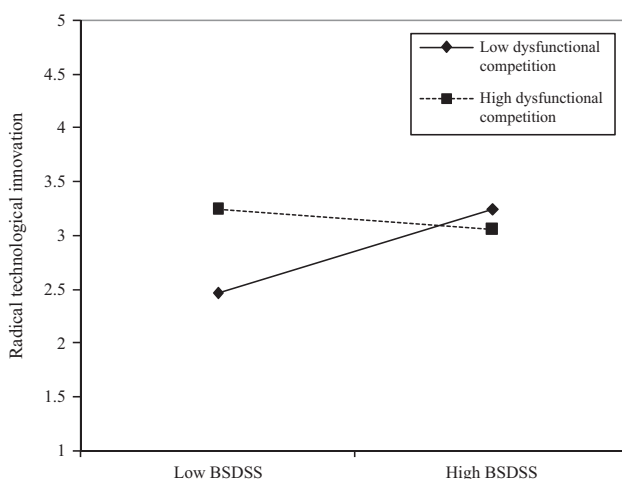


Figure 1.
The moderating effect
of dysfunctional
competition

Figure 2.
The moderating effect
of political ties

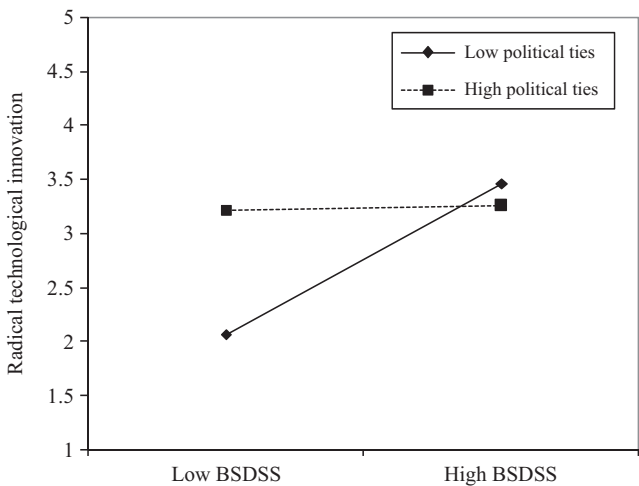
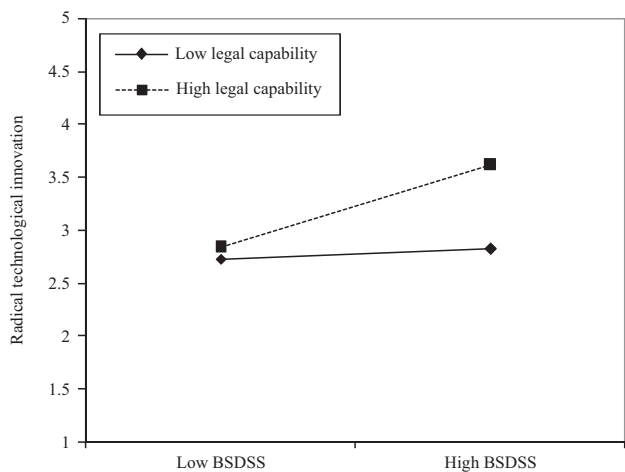


Figure 3.
The moderating effect
of legal capability



as dysfunctional competition increases, supporting the findings of model 3 that dysfunctional competition negatively moderates the relationship between BSDSS and radical technological innovations.

Discussion

Theoretical contributions

This study investigates whether, how, and when BSDSS will drive firms' radical technological innovations in Chinese market. Specifically, we find that BSDSS is positively related with firms' radical technological innovations in China. Also, such relationship will be moderated by innovation appropriability. The findings contribute to literatures on demand-side approach, innovation search and innovation appropriability.

First, we advance the emerging demand-side approach on innovations (Priem *et al.*, 2012). Boundary-spanning search for external knowledge is critical for innovations. Given the

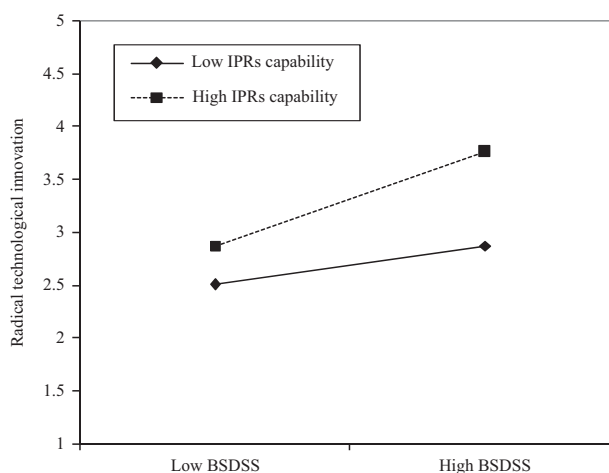


Figure 4.
The moderating effect
of IPRs capability

dominance of technology-push explanations, as well as the debate on the role of market knowledge in innovations, especially in radical technological innovations (Christensen and Bower, 1996; Slater and Narver, 1998), little research, to our best knowledge, has examined the effect of BSDSS on radical technological innovations. This study deeply explores whether and how BSDSS will fuel firms' radical technological innovations in China. By identifying BSDSS as an important antecedent for radical technological innovations, this study expands current innovation literature's horizon from supply side search to demand-side search, and thus resonates with the demand-side approach on innovations (Priem *et al.*, 2012).

Second, we establish a theoretical framework that integrates innovation search, innovation appropriability, and radical technological innovations. Previous studies have focused on how appropriability will facilitate firms to obtain benefits from their innovations (Hurmelinna *et al.*, 2007; Pisano, 2006; Teece, 1986). Our findings highlight the moderating role of appropriability in influencing firms' decisions to leverage BSDSS for radical technological innovations. Specifically, we categorize innovation appropriability into two types: innovation appropriability related with external institution environments and innovation appropriability related with internal capabilities (Teece, 1986). The empirical results indicate that in China's transitional economy, both external institutional environments and internal capabilities, as significant contingencies, will moderate the link between BSDSS and radical technological innovations. In particular, we find that internal capabilities including legal capability and IPRs capability positively moderate the main effect, whereas external institutional environments including dysfunctional competition and political ties negatively moderate the main effect. A plausible explanation for the opposite effect of political ties may be that government in China has important impact on firms' growth and success, and political ties provide firms with unique advantage such as valuable resources and policy support, which may decrease firms' entrepreneurship to be involved in radical technological innovations. Overall, this study improves our understandings over the functions of innovation appropriability.

Third, our findings also advance the research on radical technological innovations in emerging economies. In this study, considering the weak technology position and huge market opportunities, we highlight radical technological innovations in emerging economies like China can be pelled by market demand, and BSDSS can serve as a driver of radical technological innovations. In addition, we add to the literature by showing that innovation

appropriability also affects radical technological innovations in China. Because comparing with developed economies, appropriability hazards are more salient in China, which will hurt the innovation benefits, appropriability will significantly affect Chinese firms' enthusiasm in actively leveraging market opportunities to develop radical technological innovations.

Managerial implications

This study has robust practical implications for both Chinese firms and foreign firms. First, the rapid economic growth and the continual increase of income in China have spurred growing need for new products with advanced technologies. This promotes Chinese firms to develop radical technological innovations. However, as the world factory, most Chinese firms have relatively weak technological base, and have long anchored their attention on efficiency and low cost. Developing radical technological innovations will boost Chinese firms' competitive advantage in market, but how? Our findings show that BSDSS can significantly drive Chinese firms' radical technological innovations. This finding is especially meaningful for Chinese firms, because their biggest advantage is the huge market opportunity. One effective, perhaps the most effective, way for Chinese firms boosting radical technological innovations is to actively engage in BSDSS, and then drive radical technological innovations based on knowledge derived from market. In short, we suggest Chinese firms listen to the market as much as they can if they want to strengthen competitive advantages through radical technological innovations.

Second, this study emphasizes innovation appropriability will moderate the link between BSDSS and radical technological innovations in China. Appropriability can influence Chinese firms' confidence and motivation to response to market opportunities for radical technological innovations. Appropriability hazards are more salient in Chinese market where lacks well-established legal frameworks. We find that both dysfunctional competition and political ties play negative moderating roles. Thus, to effectively implement the national strategy of innovation and entrepreneurship, Chinese Government should improve intellectual property laws and protect innovations, which will boost Chinese firms' enthusiasm to achieve radical technological innovations.

In addition, our findings also have practical implications for foreign firms, especially those that value Chinese market. Foreign firms competing in Chinese market have to be aware that its unique environments will affect radical technological innovation process. For instance, the formal regulations for IPRs are still at nascent stage, so foreign firms who plan to implement radical technological innovations have to prepare for lawsuits against their local competitors. In addition, dysfunction competition will damage the effectiveness of foreign firms' radical technological innovations. For foreign firms, to win the competition in Chinese market, not only they have to acquire cutting edge technologies, but also they have to own proper knowledge on Chinese unique market rules, institutional environments, and even traditional cultures, such as competition rules, IPRs regulations, and guanxi with governmental officials. In short, we suggest foreign firms engaged in radical technological innovations in Chinese market pay more attention to its unique institutions, which will facilitate their technology innovations and strengthen competitive advantages.

Limitations and future research

This study has several limitations. First, our findings are limited to Chinese high-tech firms. They should be viewed cautiously when generalizing to other contexts, such as firms in developed economies. Future research should also collect data from other contexts to examine and correct our research findings. Second, the cross-sectional data used by our study may cast doubt on causal statements derived from the findings. A longitudinal

approach should be taken in future studies. Third, some variables, such as market capability and technology capability, may play important roles as well, and they need to be examined in further investigations.

Conclusion

We propose and examine the effects of BSDSS on radical technological innovations, as well as the moderating effects of innovation appropriability in China. Our findings show that BSDSS has a positive effect on radical technological innovations. Further, we find that dysfunctional competition and political ties as external-institution-related appropriability negatively moderate the main effect, whereas firms' legal and IPRs protection capabilities as internal-capability-related appropriability positively moderate the main effect. These findings contribute to our understanding over the joint effect of BSDSS and innovation appropriability in radical technological innovations. These above findings bear critical implications for both researchers and practitioners.

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